



Zeest: An Agentic Medical Copilot

Project Proposal

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Abstract

Patient information is fragmented across years of consultations, laboratories, imaging, and prescriptions, while existing AI tools process isolated snapshots and manual documentation consumes physician time. Zeest is a doctor-in-the-loop multi-agent clinical decision-support system centred on longitudinal patient memory. It integrates existing well-validated diagnostic models and knowledge bases rather than building from scratch, contributing novelty in real-time voice transcription, memory management, multi-agent orchestration, evidence retrieval, and safety review. Specialised agents produce evidence-grounded proposed diagnoses for doctor-in-the-loop physician review; every proposal requires explicit approval before entering the patient record.

1. Introduction

Healthcare records are scattered across encounters, systems, and formats, preventing physicians from reasoning over a complete patient trajectory. Incomplete longitudinal access drives medical errors, adverse drug events, and redundant investigations. Existing clinical decision support tools offer reference material or passive storage but no active reasoning over history. Large language models achieve strong benchmark scores yet operate in isolated sessions without persistent memory and with documented hallucination risks. Imaging and laboratory AI excels on narrow tasks but remains disconnected from patient history. Manual documentation further reduces consultation quality and time for patient care.

Zeest closes this gap. Rather than rebuilding proven diagnostic capability, it integrates pre-trained validated models and established knowledge bases as its foundation. Real-time voice transcription converts the doctor-patient exchange into structured clinical notes automatically. A multi-agent pipeline connects these inputs to persistent longitudinal memory, grounds the resulting doctor-in-the-loop diagnosis proposal in retrieved evidence, validates it through a critic agent, and presents the plan to the physician for approval, modification, or rejection. The evaluation asks: does longitudinal memory improve recommendation quality; does multi-agent reasoning outperform a single large language model; does the critic agent reduce unsupported outputs; and does multimodal integration improve decisions over text-only inputs.

2. Goals and Objectives

The goal is to build a doctor-in-the-loop agentic multimodal clinical decision-support system that maintains longitudinal patient memory, integrates real-time transcribed consultation audio with multimodal clinical data, retrieves grounded evidence, and generates safe, reviewable proposed diagnoses under continuous physician control. An ablation study will confirm that the full system outperforms a single large language model baseline. Specific objectives are:

- Hybrid longitudinal memory layer combining relational database and semantic vector store.
- Real-time voice transcription of consultations into structured clinical notes.
- Patient Memory, Clinical Case, Imaging, Laboratory, and Medication Safety agents.
- Medical Imaging Agent integrating a pre-trained validated classification model.
- Evidence and RAG Agent retrieving relevant clinical guidelines with citations.
- Clinical Reasoning Agent producing a proposed diagnosis for doctor-in-the-loop review.
- Safety and Critic Agent reviewing the proposal before physician presentation.
- Doctor-facing interface with approve, modify, and reject controls and full audit log.
- Ablation framework: single LLM vs. memory-augmented vs. full Zeest system.

3. Scope of the Project

In scope: real-time voice transcription; longitudinal hybrid memory; the full agent pipeline; multimodal data processing including imaging, labs, and transcribed notes; proposed diagnoses and treatment plans subject to doctor-in-the-loop physician approval before recording; de-identified or synthetic data. Clinical domain coverage is determined by available public datasets and validated pre-trained models; where both exist, that domain is in scope. The project's contribution is the integration, memory, reasoning, and safety layers built above existing validated diagnostic groundwork.

Out of scope: training diagnostic models from scratch; autonomous prescribing or treatment; real-world deployment or regulatory approval; domains lacking reliable datasets; native mobile apps; patient-facing conversational interfaces; the optional knowledge graph extension; emergency and ICU systems.

4. Initial Study and Work Done So Far

Rule-based CDSS tools Isabel DDx [9], UpToDate [10], and OpenMRS [11] lack active longitudinal reasoning. Med-PaLM 2 [3] and GPT-4 [4] achieved expert-level medical performance but operate in isolated sessions with no persistent memory and documented hallucination risks [3][4]. ClinicalBERT [5] and BioGPT [6] improve clinical NLP yet lack longitudinal context. Imaging AI has advanced substantially: CheXNet [7] reached radiologist-level chest X-ray classification; CheXpert [8] established benchmarks for fourteen findings; BioViL-T [12] introduced joint vision-language radiology reasoning [7][8][12]; these represent the validated groundwork Zeest integrates. ReAct [13], LangGraph [15], and Google ADK [16] provide agentic orchestration but lack patient memory and clinical safety design. RAG [17] grounds outputs in external knowledge; MedRAG [18] and MedCPT [19] show its medical effectiveness, yet none integrates RAG with patient-specific memory [17][18][19]. MIMIC-IV [20], MIMIC-CXR [21], temporal EHR work by Huang et al. [22], and Synthea [23] underpin the dataset strategy.

No existing system unifies longitudinal memory, multi-agent reasoning, multimodal integration [7][8][12], RAG-based evidence retrieval [17][18], a critic agent, and a doctor-in-the-loop approval loop. LLM systems [3][4] use single-turn sessions; imaging AI [7][8] is isolated; RAG tools [17][18] are not patient-specific. Zeest addresses each gap. Work completed: full literature review; system architecture specified; imaging candidates from CheXpert [8] and NIH data [7] identified; drug interaction knowledge bases evaluated; datasets MIMIC-IV [20], MIMIC-CXR [21], PubMed selected; PhysioNet access registered; stack confirmed: FastAPI, Next.js, LangGraph [15], Google ADK [16], Gemini, PyTorch, PostgreSQL, pgvector; Synthea [23] case outlines drafted.

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